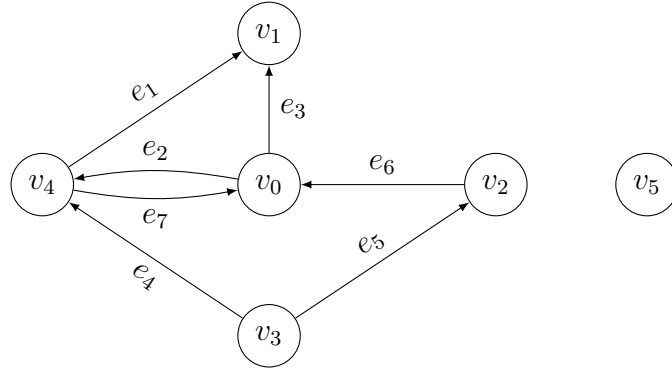


Paths

Michel Bierlaire

Solution of the practice quiz



1. An example of a path from v_1 to v_2 could be $v_1 \leftarrow v_0 \leftarrow v_2$. Another example would be $v_1 \leftarrow v_4 \rightarrow v_0 \rightarrow v_4 \leftarrow v_3 \rightarrow v_2$.
2. None of the examples above is a forward path.
3. Since the outdegree of v_1 is $d_1^+ = 0$, there is no forward path starting at v_1 .
4. There are 6 simple paths from v_3 to v_1 :

- $\mathcal{P}_1 : v_3 \rightarrow v_4 \rightarrow v_1$, which is a simple forward path:

$$[\mathcal{P}_1^{\rightarrow} = \{e_4, e_1\}, \mathcal{P}_1^{\leftarrow} = \emptyset],$$

where $\mathcal{P}_1^{\rightarrow}$ is the set of forward arcs of the path $\mathcal{P}_1$ and $\mathcal{P}_1^{\leftarrow}$ is the set of backward arcs.

- $\mathcal{P}_2 : v_3 \rightarrow v_4 \rightarrow v_0 \rightarrow v_1$, which is a simple forward path:

$$[\mathcal{P}_2^{\rightarrow} = \{e_4, e_7, e_3\}, \mathcal{P}_2^{\leftarrow} = \emptyset].$$

- $\mathcal{P}_3 : v_3 \rightarrow v_4 \leftarrow v_0 \rightarrow v_1$, which is a simple path:

$$[\mathcal{P}_3^{\rightarrow} = \{e_4, e_3\}, \mathcal{P}_3^{\leftarrow} = \{e_2\}].$$

- $\mathcal{P}_4 =: v_3 \rightarrow v_2 \rightarrow v_0 \rightarrow v_1$, which is a simple forward path:

$$[\mathcal{P}_4^{\rightarrow} = \{e_5, e_6, e_3\}, \mathcal{P}_4^{\leftarrow} = \emptyset].$$

- $\mathcal{P}_5 =: v_3 \rightarrow v_2 \rightarrow v_0 \rightarrow v_4 \rightarrow v_1$, which is a simple forward path:

$$[\mathcal{P}_5^{\rightarrow} = \{e_5, e_6, e_2, e_1\}, \mathcal{P}_5^{\leftarrow} = \emptyset].$$

- $\mathcal{P}_6 =: v_3 \rightarrow v_2 \rightarrow v_0 \leftarrow v_4 \rightarrow v_1$, which is a simple path:

$$[\mathcal{P}_6^{\rightarrow} = \{e_5, e_6, e_1\}, \mathcal{P}_6^{\leftarrow} = \{e_7\}].$$

5. The path $v_3 \rightarrow v_4 \rightarrow v_0 \rightarrow v_4 \rightarrow v_1$ is not simple since the node v_4 is repeated.
6. The path $v_1 \leftarrow v_0 \leftarrow v_2 \leftarrow v_3 \rightarrow v_4 \rightarrow v_1$ is an example of a simple cycle since the origin and the destination coincide, and no other node is repeated. Another one would be $v_0 \rightarrow v_4 \rightarrow v_0$.
7. The path $v_1 \leftarrow v_0 \rightarrow v_4 \rightarrow v_0 \rightarrow v_1$ is a cycle since the origin and the destination coincide. It is not a simple cycle as node v_0 appears twice in the path.